

Scalability & Future Plan

Date	12 July 2026
Team ID	SWTID-2026-8977
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports a limited number of concurrent users in the current deployment.	Performance may decrease during high user traffic.	High
2	Uses a lightweight local database (SQLite) for storing user data and learning history.	Limited scalability for large datasets and multiple users.	Medium
3	Limited personalization and progress tracking features.	Users cannot monitor long-term learning progress or receive advanced recommendations.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports individual and small-scale users.	Deploy the application on cloud platforms (Google Cloud, AWS, or Azure) with load balancing to support thousands of concurrent users.
2	Data Storage	SQLite database for local storage.	Upgrade to PostgreSQL or MongoDB with cloud storage for improved scalability and reliability.
3	Performance	FastAPI with Google Gemini API provides good response times for moderate workloads.	Implement caching, asynchronous request processing, and API optimization to improve performance under heavy loads.

4	Security	API key protection and input validation are implemented.	Add user authentication, role-based access control (RBAC), HTTPS enforcement, encrypted data storage, and OAuth 2.0 authentication.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Login, Profile Management, Learning Progress Tracking, and Quiz Score Analytics	3–6 Months	Enables personalized learning experiences and progress monitoring.
Phase 3	Voice-Based Learning Assistant, PDF Upload & Summarization, Multi-Language Support, AI Flashcards	6–9 Months	Makes the platform more interactive, accessible, and suitable for diverse learners.
Phase 4	Mobile Application (Android & iOS), Real-Time Collaboration, AI Tutor Chatbot, Cloud Deployment, Learning Analytics Dashboard	9–12 Months	Expands accessibility, supports large-scale educational use, and enhances user engagement through advanced AI capabilities.